

Chihyeong Lee | Curriculum Vitae

Seoul National University – Seoul, Korea

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Google Scholar Profile

Research Interests

- Biomechanics, Musculoskeletal Simulation, Reinforcement Learning, Imitation Learning, Machine Learning & Deep Learning, Wearable Robotics, Rehabilitation Engineering, Gait Performance

Education

Seoul National University **Seoul, Korea**

Ph.D. Candidate in Sport Science, GPA: 3.98/4.00

Mar 2023 – June 2025

Advisor: Prof. Joeeun Ahn

Department of Health and Human Performance, University of Houston **Texas, USA**

Visiting Scholar

July 2024

Seoul National University **Seoul, Korea**

Master of Sport Science, GPA: 3.91/4.00

Mar 2021 – Feb 2023

Advisor: Prof. Joeeun Ahn

Seoul National University **Seoul, Korea**

Bachelor of Sport Science, GPA: 3.79/4.00

Mar 2014 – Feb 2021

Publications

Peer-reviewed Journal Articles

- Park SH, **Lee C**, Park H, Ahn J, Lee B. "Independent and synergistic effects of visual biofeedback and phase-specific belt deceleration on affected-leg propulsion in post-stroke split-belt treadmill training." *Journal of NeuroEngineering and Rehabilitation*, 2025 (Under review)
- Hong T¹, **Lee C**¹, Chang, S, Choi E, Kim B, Ahn J, Park Y. "Design of A Fully-Soft Lift-Assist Wearable Suit Powered by Flat Inflatable Artificial Muscles." *IEEE Robotics and Automation Letters*. 2025 Feb 13; IEEE. (IF: 4.6) (¹Contributed equally) **[Read Full Paper]**
- **Lee C**, Ahn J, Lee B. "The effects of perturbation intensities on backward slip-falls induced by a split-belt treadmill." *Scientific Reports*. 2025 Feb 11; 15(1):5108. (IF: 3.8) **[Read Full Paper]**
- **Lee C**, Ahn J, Lee B. "A systematic review of the long-term effects of using smartphone-and tablet-based rehabilitation technology for balance and gait training and exercise programs." *Bio-engineering*. 2023 Sep 28; 10(10):1142. (IF: 3.8) **[Read Full Paper]**

Peer-reviewed Conference Papers

- **Lee C**, Kim H, Kim C, Ahn J, Kim K, Kwon Y. "Evaluation of Kinematic Similarity between Non-Assisted and Assisted Walking with a Hip Exoskeleton Using a Vector Coding Technique ." *47th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*, July 14–17, 2025.

- **Lee C**, Ahn J, Lee B. "Evaluating the Impact of Different Slipping Velocities on Falls and Non-Falls Induced by a Programmable Split-Belt Treadmill." *47th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*, July 14–17, 2025.
- Yoo D, **Lee C**, Ahn J, Lee B. "Age-related adaptation of the body's kinematic responses to unpredictable trip perturbations induced by a split-belt treadmill." *45th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*, July 24–27, 2023. **[Read Paper]**

Conference Abstracts.....

- **Lee C**, Ahn J, Lee B. "Referent-Configuration Imitation Learning (RCIL) for Predictive Musculoskeletal Simulation: Faster and more stable gait imitation by controlling referent configurations instead of muscle activations." *10th World Congress of Biomechanics (WCB)*, July 11–15, 2026.
- Rivera A, Park H, **Lee C**, Ahn J, Park SH, Lee B. "Enhancing paretic propulsion post-stroke via real-time visual biofeedback and adaptive dual-belt treadmill control." *55th Annual Meeting of the Society for Neuroscience (SfN)*, November 15–19, 2025.
- **Lee C**, Choi E, Kim M, Ahn J. "Downhill Walking Classification for Gait-Assive Wearable Robot: A Machine Learning and Deep Learning Approach." *49th Annual Meeting of the American Society of Biomechanics (ASB)*, August 13–16, 2025.
- **Lee C**, Ahn J, Lee B. "Effects of slip velocity and duration on fall incidence: Evidence from unexpected slip perturbations induced by a split-belt treadmill." *49th Annual Meeting of the American Society of Biomechanics (ASB)*, August 13–16, 2025.
- **Lee C**, Kim H, Kim C, Ahn J, Kim K, Kwon Y. "KINEMATIC COUPLING RELATIONSHIPS BETWEEN HIP AND KNEE JOINTS WHEN WALKING WITH A HIP EXOSKELETON." *49th Annual Meeting of the American Society of Biomechanics (ASB)*, August 13–16, 2025.
- **Lee C**, Choi E, Choi Y, Kim C, Lee J, Min H, Ahn J. "Gait-Assistive Wearable Robots: Biomechanical Evaluation Schema." *2025 Soft Robot Research Center (SRRC) Symposium*, Feb 21, 2025.
- **Lee C**, Choi E, Min H, Ahn J. "Quantitative Evaluation of Familiarization with Walking Assistive Wearable Robots - A preliminary study on hip joint assistive robots." *2024 Korea Society of Sport Biomechanics (KSB)*, Nov 21, 2024.
- **Lee C**, Ahn J. "REAL-TIME BACKWARD SLIP DETECTION USING A SLIP-INDUCING SYSTEM AND MACHINE LEARNING METHODS." *41st International Society of Biomechanics in Sports (ISBS)*, July 12–16, 2023. **[Read Abstract]**
- Yoo D, **Lee C**, Ahn J, Lee B. "ASSOCIATION BETWEEN AGE AND BODY'S KINEMATIC RESPONSES TO UNPREDICTABLE GAIT PERTURBATION." *41st International Society of Biomechanics in Sports (ISBS)*, July 12–16, 2023. **[Read Abstract]**

Projects

- **Development of fabric-type hybrid actuators with embedded soft sensors and wearable robotic technologies for industrial workers.**
 - Funding: Korea Evaluation Institute of Industrial Technology (KEIT), funded by the Ministry of Trade, Industry and Energy (MOTIE), Korea (Grant No. 20008912)

- *Period: Mar. 2021 – Dec. 2023*
 - *Project Budget: KRW 2,815,584,000 (~USD 2.0M)*
 - *Role: Developing lightweight and flexible wearable robots to assist lifting tasks in industrial settings through various structural designs and support mechanisms, and collaborating with a multidisciplinary team to integrate biomechanics and robotics for improving efficiency and safety in industrial work environments.*
- **Development of wearable robot products and service platforms for healthcare in home and daily-life environments.**
 - *Funding: Korea Evaluation Institute of Industrial Technology (KEIT), funded by the Ministry of Trade, Industry and Energy (MOTIE), Korea (Grant No. 20018157)*
 - *Period: Mar. 2022 – Dec. 2025*
 - *Project Budget: KRW 8,455,600,000 (~USD 6.0M)*
 - *Role: Evaluating wearable robots for lift-assist and walking support in home and daily living environments, and providing feedback for development improvement while collaborating with a multidisciplinary team to integrate biomechanics and robotics for enhancing efficiency and safety in home and daily living environments.*
- **Development of an advanced fall-inducing system for fall detection and prevention.**
 - *Funding: National Research Foundation of Korea (NRF), funded by the Ministry of Science and ICT (MSIT), Korea (Grant No. RS2023-00208052)*
 - *Period: Dec. 2022 – Nov. 2023*
 - *Project Budget: KRW 151,132,000 (~USD 110K)*
 - *Role: Developing an advanced fall-inducing system and validating its performance through clinical assessment for fall detection and prevention.*
- **Development of clinical and biomechanical evaluation metrics for flexible wearable care robots.**
 - *Funding: Korea Health Industry Development Institute (KHIDI), funded by the Ministry of Health and Welfare (MOHW), Korea (Grant No. HK23C0071)*
 - *Period: Mar. 2023 – Dec. 2027*
 - *Project Budget: KRW 1,440,000,000 (~USD 1.0M)*
 - *Role: Evaluating walking assistive wearable robots and developing standardized assessment protocols. Collaborating with a multidisciplinary team to integrate biomechanics, robotics, and clinical assessment for the development of reliable and valid evaluation standards.*
- **Development and clinical evaluation of a patient-tailored intelligent gait rehabilitation system using real-time propulsion assistance and visual feedback.**
 - *Funding: National Research Foundation of Korea (NRF), funded by the Ministry of Science and ICT (MSIT), Korea (Grant No. RS2024-00446461)*
 - *Period: Mar. 2024 – Dec. 2026*
 - *Project Budget: KRW 383,535,000 (~USD 280K)*
 - *Role: Developing a patient-tailored intelligent gait rehabilitation system to improve propulsion in post-stroke patients, and validating the system through clinical trials.*

Invited Lecture

- **Lee C.** "Advancing Laboratory Fall Research - Toward Fall Prediction and Fall-Prevention Training." Guest Lecture in *Biomechanics of Lower Extremities* course, Seoul National University, Seoul, Korea, Nov 2024.

Other Experience

Seoul National University, 2nd Division Team	Seoul, Korea
Varsity Basketball Coach	Sep 2020 – 2022, 2025
- 3rd Place, National University Basketball Tournament	2022
Varsity Basketball Player	May 2014 – Aug 2019
- 3rd Place, National University Basketball Tournament	2014, 2017, 2019
- 3rd Place, National University Basketball League	2014
Jeju National Hospital	Jeju, Korea
- Social Service Personnel (Mandatory Military Service)	May 2018 – Aug 2019
Gangneung Public Health Center	Gangneung, Korea
- Social Service Personnel (Mandatory Military Service)	Sep 2017 – Apr 2018

References

Prof. Jooeun Ahn

Associate Professor, Department of Kinesiology, Seoul National University, Seoul, Korea

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Profiles: [SNU Personal Page](#) | [Google Scholar](#)

Prof. Beom-Chan Lee

Associate Professor, Department of Health and Human Performance, University of Houston, Texas, USA

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Prof. Siddhartha Bikram Pandey

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